Unit 4 · Outcome 2 · Energy Sources

Keeping the lights on: base and peak load

KK09 VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.

Electricity can’t be stockpiled — supply must equal demand every single second

01 The balancing act Electricity can’t be stockpiled — supply must equal demand every single second Coal, gas, water and sunshine can be stored. The electricity made from them, mostly, cannot. The amount a grid generates has to match the amount everyone is using at that instant , or the system’s frequency drifts away from 50 Hz and protective relays start tripping — that’s a blackout. Analogy — the tank with no tank Picture a water tower that supplies a town but has no actual storage tank — water flows straight from the pumps to every tap. If everyone

Reading the demand curve

02 Base vs peak load Reading the demand curve Over any 24 hours, total electricity demand rises and falls in a familiar pattern: low overnight, a morning lift, a dip through the working day, then a sharp evening peak as people get home, cook, heat or cool, and switch everything on at once. interactive A day on the Yarrabin grid · hover the curve · the shaded floor is base load, the crest is peak load Your browser blocked the canvas. Base load = the lowest point the demand curve ever reaches (always-on demand). Peak load = the highest point (the evening c

Dispatchability: can you turn it up when you need it?

03 Matching sources to load Dispatchability: can you turn it up when you need it? The single property that decides whether a source suits base or peak load is dispatchability — whether grid operators can raise or lower its output on command . This splits every source into three groups. Role Behaviour Best-suited sources Why Base-load Runs continuously at a steady output; slow to start up or shut down Black/brown coal , nuclear , large hydro , geothermal , biomass Cheap to keep running once hot; high capacity factor (~80–90%); inefficient and costly to ke

Role	Behaviour	Best-suited sources	Why
Base-load	Runs continuously at a steady output; slow to start up or shut down	Black/brown coal , nuclear , large hydro , geothermal , biomass	Cheap to keep running once hot; high capacity factor (~80–90%); inefficient and costly to keep ramping
Peaking	Dispatchable — starts, ramps and stops quickly	Open-cycle gas turbines, hydro (esp. pumped hydro), grid batteries	Fast response fills short spikes; higher cost per unit, so only run during peaks
Variable (intermittent)	Output set by weather/time, not by the operator — must-take	Solar (midday only), wind (when it blows)	Zero fuel cost & emissions, but can’t be “turned up” for the evening peak; needs firming

Run the Yarrabin grid for a day

04 Grid dispatch simulator Run the Yarrabin grid for a day This is the dot point made tangible. The black line is the real demand curve from §02. Your job: build a supply stack beneath it that meets demand at every hour — without big shortfalls (blackouts) or huge surpluses (wasted, curtailed power). Adjust each generator and watch the stack and the gauges respond. Yarrabin Grid · 24-hour dispatch stack must reach the demand line at all 24 hours Old grid: coal only Solar + wind only Balanced firmed mix Clear all Inspect hour 18:00 Add some generation t

Why more solar makes the evening peak harder

05 The duck curve Why more solar makes the evening peak harder Here's the twist that makes this dot point genuinely current. Rooftop and farm solar don't reduce demand evenly — they slash it only in the middle of the day . What the grid's big generators actually have to supply is net demand = total demand − solar. As solar grows, that net-demand curve develops a deep midday belly and a steep evening neck. Plotted out, it looks like a duck. 3D · drag Net demand ribbon — drag the solar slider below. More solar deepens the midday belly and steepens the even

From "current" to "projected" — what the VCAA wording means

06 Current & projected needs From "current" to "projected" — what the VCAA wording means The dot point asks about current and projected needs. These describe two different grids. Today's grid (current) Base load still leans on ageing coal (e.g. Loy Yang in the Latrobe Valley) and large hydro . Peaks met by gas turbines and hydro . A fast-growing slice of variable solar and wind, plus the world's highest rooftop-solar uptake. Tomorrow's grid (projected) Coal stations retire — the base-load floor must be held a new way. Far more variable renewables → a dee

How base and peak needs are met — at two levels

07 Individual & societal How base and peak needs are met — at two levels The dot point ends with " at individual and societal levels ." Meeting demand isn't only an engineering job for big operators; households shape the curve too. A strong answer addresses both . Societal level (the grid) Generation mix — build base-load + peaking + variable + storage so the stack can meet any hour. Dispatch & markets — AEMO/the NEM forecast demand and dispatch the cheapest available generators in real time. Storage & firming — pumped hydro and grid batteries store m

Decode what the question is really asking

◈ **COMMAND WORDS**

08 Command words Decode what the question is really asking The verb sets the marks. Tap a command word to see what an examiner expects on this dot point.

Six ways students lose marks here

09 Exam traps Six ways students lose marks here 1 · "Base load = the most energy" No. Base load is the minimum continuous demand (the floor), not the largest amount of energy. Define it as a level of demand , not a quantity of supply. 2 · "Solar/wind can provide base load" Alone, no — they're variable , not continuous or dispatchable. Say "solar + storage can help meet base load," never solar by itself. 3 · Confusing capacity with energy A 100 MW wind farm rarely delivers 100 MW. Distinguish installed capacity (MW) from delivered energy (MWh) via capacit

Write a paragraph that earns the marks

10 P·E·L trainer Write a paragraph that earns the marks VCE extended responses reward structure. Use P·E·L : make a Point , support it with Evidence (a named source, figure or mechanism), then Link back to the question. Type a response below — the meters detect whether each element is present and flag key terminology. Practice prompt: Explain why solar power, despite its growth, cannot on its own meet a grid's base and peak load needs. (4 marks) Point 0% A clear claim answering the verb Evidence 0% Mechanism, source, or figure Link 0% Ties back to base/p

Same question, two responses

✗ WEAK ANSWER

weak response (1–2/4) ✗ Weak — 1.5/4 Coal is better than batteries because it makes a lot of electricity all the time. Batteries are renewable and good for the environment but they don't make as much power . So coal is more reliable but batteries are cleaner.

✓ STRONG ANSWER & MARKING

Why it loses marks: no base/peak framing; "batteries are renewable" is wrong (they're storage, not a sour

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Five graded questions · 20 marks

★ PRACTICE & SELF-MARK

12 Practice questions Five graded questions · 20 marks Attempt each in writing first, then reveal the marking guide and self-mark. Your score feeds the dock in the corner.

If you remember nothing else

✦ RECAP / CHEAT SHEET

13 Key takeaways If you remember nothing else Supply must match demand every second. Electricity is barely storable, so the 24-hour demand curve drives everything. Base load = the continuous minimum demand (the floor); peak load = the brief maximum (the evening/heatwave crest). They're parts of the demand curve, not types of source. Dispatchability decides the match. Continuous, slow sources (coal, nuclear, geothermal, large hydro) suit base load; fast, dispatchable sources (gas, hydro, batteries) suit peaks. Variable renewables (solar, wind) supply che

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

1.5/4 Coal is better than batteries because it makes a lot of electricity all the time. Batteries are renewable and good for the environment but they don't make as much power . So coal is more reliable but batteries are cleaner. Why it loses marks: no base/peak framing; "batteries are renewable" is wrong (they're storage, not a source); "don't make as much power" confuses capacity with role; it's two separate descriptions, not a true comparison; no dispatchability or capacity-factor language.

✓ STRONG / MODEL ANSWER

Brown coal suits base load : it runs continuously at a high capacity factor (~85%), so it economically holds the always-present demand floor — but it ramps slowly, so it cannot follow short spikes, and it emits heavily. Grid batteries suit peak load : they are storage, not generation , dispatching stored energy within seconds to cover the brief evening peak, with no direct emissions — but their stored energy is limited, so they cannot sustain base load for long unrecharged. Overall , the two are complementary rather than competing: coal (or a low-emission base-load source) holds the floor while batteries firm the peak — so a reliable grid needs both roles filled, not one source. Why it score

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.